

SUMIT TIWARI

6-Month AI / ML / Data Science Roadmap

Goal: Build strong foundations in Data Science, Machine Learning, Deep Learning, Generative AI, RAG Systems, and Agentic AI while continuing DSA practice and project development.

Month 1: Data Science Foundations

- 1 Week 1: NumPy
- 2 Week 2: Pandas
- 3 Week 3: Matplotlib & Seaborn
- 4 Week 4: SQL

Project: Student Marks Analyzer, IPL Analysis, Sales Dashboard, Student Database Analysis

Month 2: Math + ML Foundations

- 1 Linear Algebra
- 2 Statistics
- 3 Probability
- 4 Scikit-Learn Workflow

Project: House Price Predictor

Month 3: Core Machine Learning

- 1 Linear Regression
- 2 Logistic Regression
- 3 KNN
- 4 Decision Trees & Random Forest

Project: Spam Classifier

Month 4: Advanced ML + Deep Learning

- 1 Feature Engineering
- 2 Hyperparameter Tuning
- 3 PCA
- 4 Neural Networks

Project: Deep Learning Foundations

Month 5: Computer Vision

- 1 PyTorch
- 2 CNNs
- 3 Image Processing
- 4 Image Classification

Project: Image Classifier

Month 6: Generative AI + RAG + Agents

- 1 Transformers
- 2 Embeddings

3 RAG

4 LangGraph, CrewAI, FastAPI, Docker

Project: Resume Analyzer, PDF Chatbot, AI Research Agent

Weekly Schedule

- 1 Monday–Friday: 45 minutes DSA
- 2 Monday–Friday: 1.5–2 hours AI/ML
- 3 Daily: 15 minutes revision and notes
- 4 Saturday: Project Building
- 5 Sunday: Finish Project + GitHub Push + Revision

Portfolio Projects Checklist

- 1 ■ Student Marks Analyzer
- 2 ■ IPL Data Analysis
- 3 ■ Sales Dashboard
- 4 ■ Student Database Analysis
- 5 ■ House Price Predictor
- 6 ■ Spam Classifier
- 7 ■ Image Classifier
- 8 ■ Resume Analyzer
- 9 ■ PDF Chatbot
- 10 ■ AI Research Agent